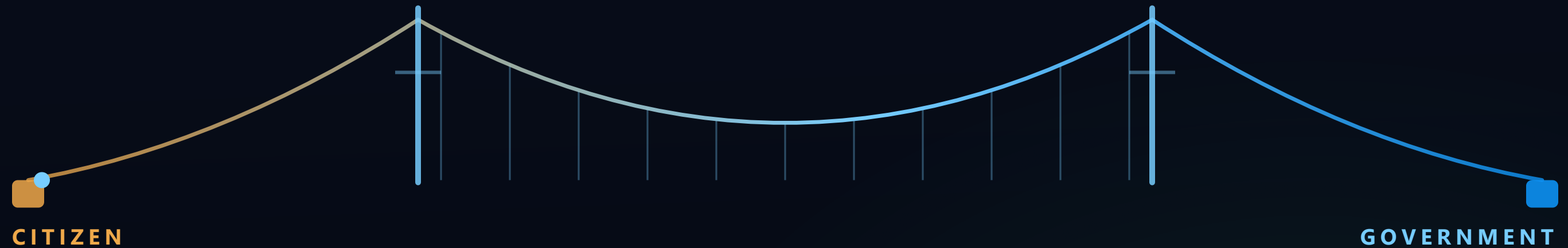


न्यायसेतु

NyayaSetu-GovAgent

Offline-First Multilingual AI for Rural Governance

Shaurya Punj · Thapar Institute of Engineering & Technology



— THE PROBLEM

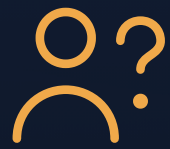
Grassroots governance breaks where the internet does.



01

Patchy connectivity

rural bandwidth · intermittent



02

Low digital literacy

forms & portals fail



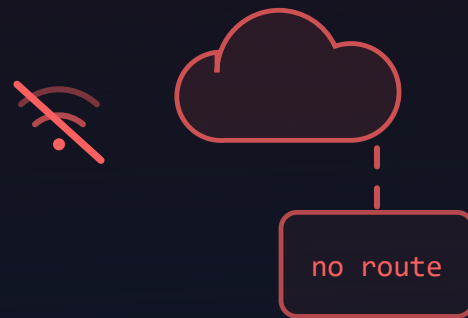
03

22 official languages

service is English-first

— THE GAP

Urban tools assume connectivity. Villages don't have it.



Cloud chatbot

Every answer needs a live network round-trip.

OFFLINE = DEAD

VS



Village kiosk

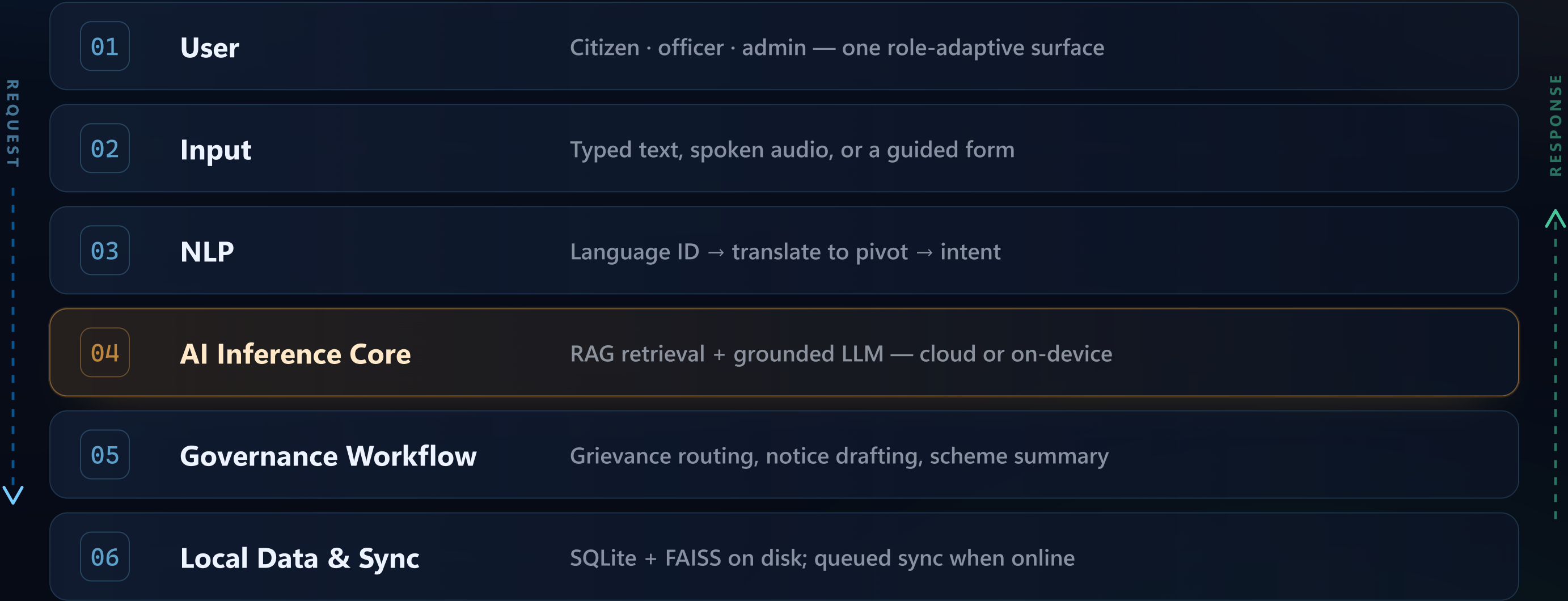
Model, index and records live on the machine.

STILL WORKS

A governance co-pilot
that works **offline**, in your language.



Six layers. Offline-first.



Retrieve real government text, then answer.



— MULTILINGUAL

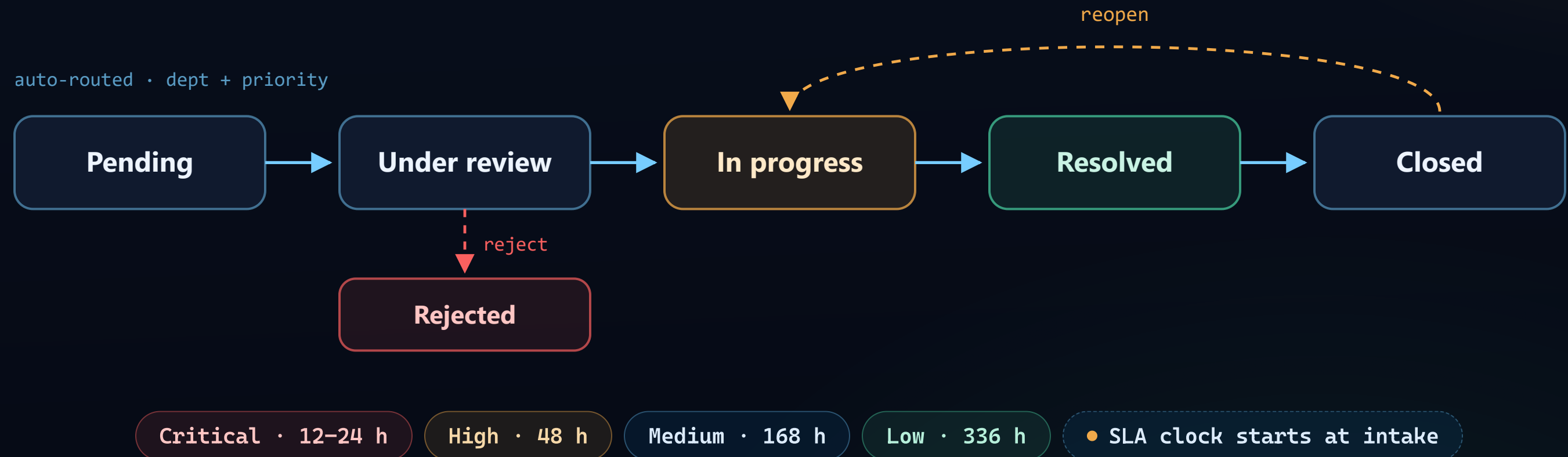
Speak the citizen's language. Twelve of them.



हिन्दी	HI	বাংলা	BN	தமிழ்	TA
తెలుగు	TE	मराठी	MR	ગુજરાતી	GU
ಕನ್ನಡ	KN	മലയാളം	ML	ਪੰਜਾਬੀ	PA
ଓଡ଼ିଆ	OR	অসমীয়া	AS	English	EN



Every complaint: routed, prioritised, deadlined, traceable.



Cloud when present. Fully on-device when not.

CLOUD

Groq LLM

Sarvam voice

Postgres

needs network



NO INTERNET → STILL ANSWERS

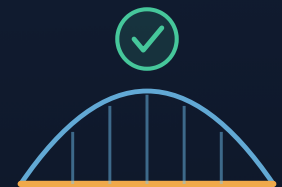
ON-DEVICE

llama.cpp · Qwen-3B

fastText LID

faster-whisper

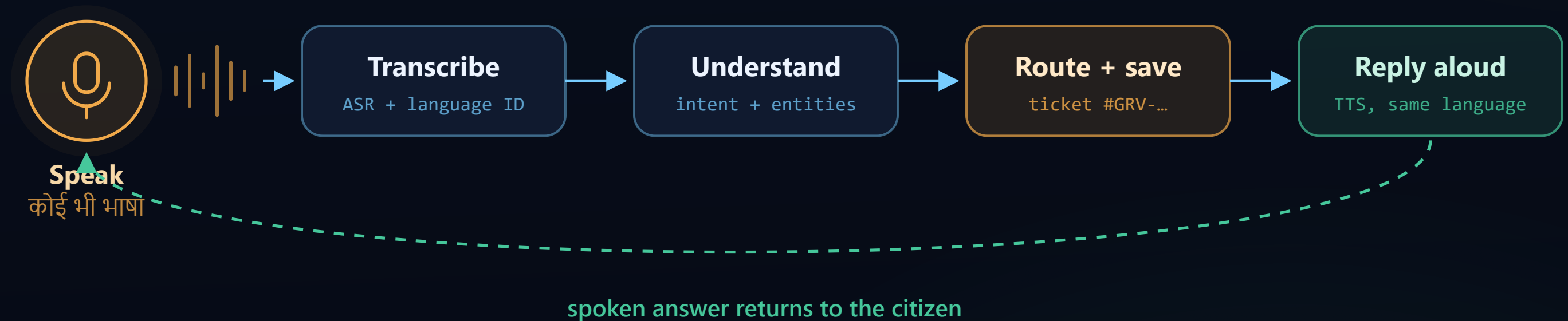
SQLite + FAISS



CRDT sync bridge

Queue locally, reconcile later — no lost tickets.

One spoken sentence. Zero typing.



— IMPLEMENTATION

One codebase. Laptop to cloud.

SERVE

- FastAPI REST
- Uvicorn ASGI
- JWT + bcrypt auth

RETRIEVE

- FAISS IndexFlatIP
- transformers 768-d
- fastText lang-ID

INFER

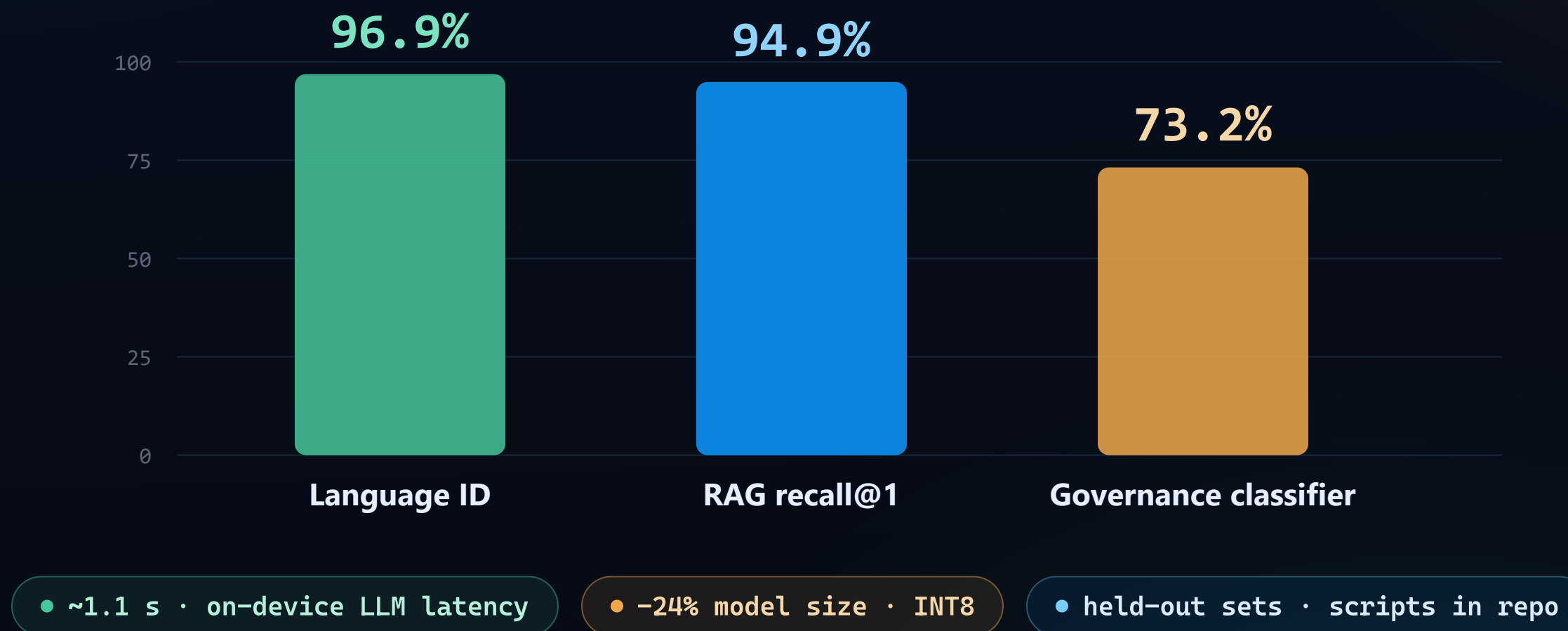
- llama.cpp GGUF
- Qwen-3B INT8
- faster-whisper ASR

REACH

- React UI
- Electron desktop
- Capacitor android

— EVALUATION

Real, reproducible measurements.



— COMPARISON

Built for the last mile.

	NYAYASETU	CLOUD CHATBOT	WEB PORTAL	MANUAL DESK
Works offline	✓	✗	✗	✓
12 Indian languages	✓	—	✗	—
Local-first privacy	✓	✗	✗	—
Usable at low literacy	✓	—	✗	✓
Traceable & instant	✓	✓	✓	✗

— LIMITATIONS & FUTURE

Honest gaps. Clear road ahead.

🕒 Corpus still growing — nightly cron

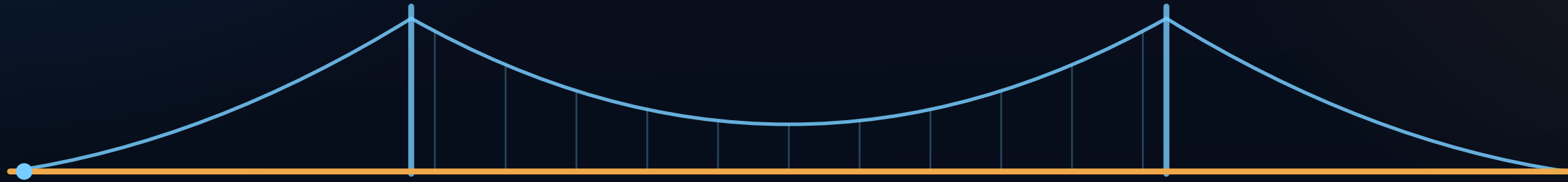
🔗 Pivot translation — BLEU below dedicated NMT

📍 No field pilot yet — lab devices only



— NEXT

- All 22 scheduled languages
- Panchayat field pilots, real hardware
- On-device continual learning



न्यायसेतु — technology that reaches the last mile.



• github.com/<org>/nyayasetu

• code • datasets • eval scripts

THANK YOU • QUESTIONS?